

Course Book

Econometrics III

B.Sc. Econometrics and Data Science, year 3

B.Sc. Econometrics and Operations Research, year 3

Period 4, Academic year 2025/26



1 Course goals

In the first part of the course, you will learn more about multivariate regression and panel data analysis, including fixed effects, two-way fixed effects and random effects approaches to panel data models and difference-in-difference methods. The second part of the course will give you a thorough introduction to the theory and practice of multiple time series analysis, including vector autoregressive, vector error correction, and dynamic factor models.

2 Teachers

- Lectures: **prof. dr. Siem Jan Koopman**, professor at the Department of Econometrics and Data Science. Email: s.j.koopman@vu.nl
- Tutorials, Q&A's, coordination of assignments: **Yannick van Etten**, PhD student at Tinbergen Institute. Email: y.j.van.etten@student.vu.nl

3 Lectures and Tutorials

3.1 Lectures

- The **lectures** are delivered in-person on Monday and Tuesday; see <http://rooster.vu.nl>.
- The slides are made available via Canvas.
- There are 6 lecture weeks during the SBE teaching period 4.
- **Note:** You are strongly encouraged to attend all lectures.

3.2 Tutorials

- The **tutorials** are delivered in-person on Wednesday, there are three sessions; see <http://rooster.vu.nl>. Your questions are addressed and solutions to various exercises are presented.
- **Note:** Your attendance is highly recommended.

3.3 Office hours for assignments

- Office hours for supporting your work on the assignments will take place, starting from week 2; see Canvas.

3.4 Self-study

Every week, you will be asked to solve some problems in self-study, prepare the practical exercises, or work on the graded assignments (see the course schedule for details). It is highly recommended to form study groups and discuss amongst each other every week. You may also use the "Discussions" section on the Canvas course page.

4 Assignments

There will be **three** assignments. The deadlines for handing them in will be communicated via the course page on Canvas. Handing in the assignments is **compulsory**. The results count for 50% (i.e. 20% for Assignment 1, 20% for Assignment 2, and 10% for Assignment 3) of the final exam grade. There will be no resits for the assignments. Assignment work of previous years will not be considered. The assignments will consist of empirical questions that need to be solved using **Python** or **Ox**, or, alternatively but not preferably, R or Matlab. You **cannot** use Stata, Eviews or gretl. The assignments are graded together with the exam. A sketch solution to the assignment questions will be provided. The assignments should be solved in **groups of four**. Every assignment will come with warm-up exercises, for which solutions will be provided.

5 Textbooks

A few suggestions:

- For the part on panel data models, we base most material on the book of Baltagi (2013). The book of Wooldridge (2010) may also be of interest.
 - Baltagi, Badi H. (2013) *Econometric Analysis of Panel Data*. Wiley.
 - Wooldridge, Jeffrey M. (2010) *Econometric Analysis of Cross Section and Panel Data*. The MIT Press.
- The Hamilton (1994) book is not directly used in this course, but it is a standard textbook in time series analysis and also contains several chapters on multiple time series models:
 - Hamilton, James D. (1994): *Time Series Analysis*. Princeton University Press.
- Other references are
 - Helmut Lutkepohl (2006): *New Introduction to Multiple Time Series Analysis*. Springer.
 - Abadir, K.M. and J.R. Magnus (2005): *Matrix Algebra*. Cambridge.

6 Content outline

The contents covered in each week will be (approximately) the following:

Week 1 Multiple equations in regression

- Multiple equations in regressions
- Matrix Algebra
- Seemingly unrelated regression equations
- Generalized least squares
- Feasible generalized least squares
- Pooled regressions

Week 2 Panel data models

- Panel Data
- Pooled Estimator

- Random effects model
- Random effects: between estimation
- Random effects: within estimation

Week 3 Panel data models, continued

- Fixed effects model
- Random effects or fixed effects?
- Hausman test
- Recent developments
- Group-means
- Two-ways fixed effects model
- Difference-in-differences method

Week 4 Vector autoregressive model of order 1

- VAR(1) model
- Wold representation
- Dynamic properties
- Forecasting
- Granger causality
- Impulse response function

Week 5 Vector autoregressive model of order p

- Parameter estimation
- Conditional regression estimation
- Maximum likelihood estimation
- Hypothesis testing
- Generalization to VAR(p) model
- What p to chose ?

Week 6 Cointegration and vector error correction model

- Non-stationary time series
- Unit roots
- Cointegration
- Vector error correction model

7 Software: Python or Ox

In the warm-up exercises, we will use the software **Python** to illustrate materials, analyze data sets, and solve exercises. If requested, **Ox** versions of the code are also available. Python is a common programming language used among data experts at universities and the industry. Ox is a matrix object-oriented programming language used in econometric research. Both platforms are open source and can be installed in any environment, used on any machine, no licensing or agreements needed. The software is under active development, improving constantly, and supported by both professional and academic developers.

- Download Python: python.org
- Download Ox Console: doornik.com

8 Exam

There will be a multiple choice exam that covers the materials we discuss in the lectures and in the tutorials. More information will be provided in due time.

9 Detailed schedule

On the Canvas page of the course (Module “Course manual”), you can also find a detailed schedule covering every day of the eight weeks of the course.